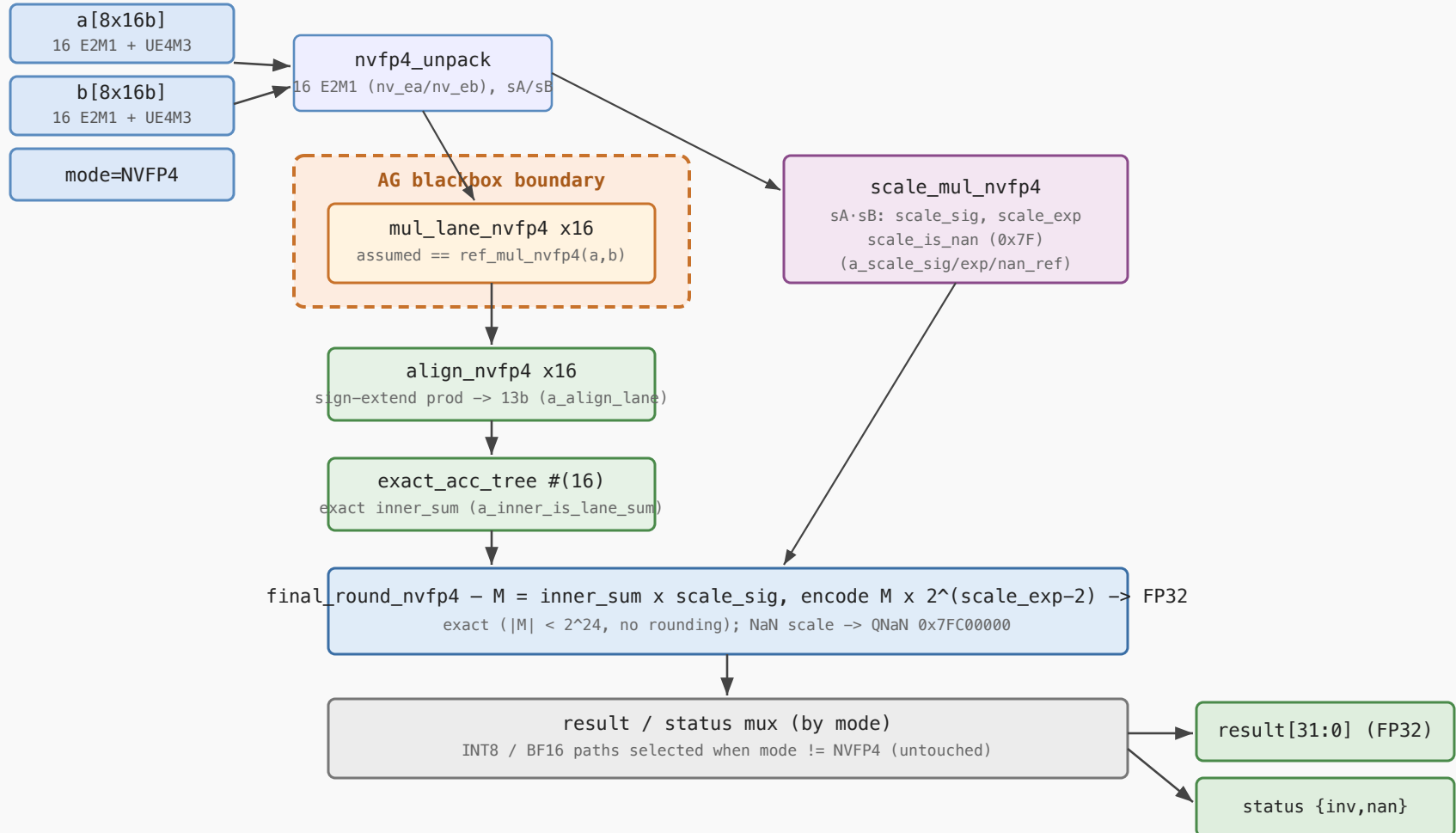


dotprod_top – NVFP4 Datapath (M4)

block-16 E2M1 + UE4M3 scale, packed in a/b[8x16b]; NVFP4_INNER_W=13; result FP32



Assume-Guarantee top proof (blackbox lane, assert over pure DUT nets)

lane: mul_lane_nvfp4 == ref_mul_nvfp4 (blackboxed above, assumed here; proven standalone)
top: a_inner_is_lane_sum (inner_sum == Sum wide[i]) + a_align_lane (wide[i]==sign_ext(prod[i]))
+ a_scale_sig/exp/nan_ref – all PURE DUT nets, no golden call in the miter
round: final_round_nvfp4 exact (directed sim vs golden)
=> result == dotprod_ref_nvfp4 by transitivity (unpack indexing DUT==golden, sign-ext widths match)